

Reason, Plan, Animate: Trajectory-Grounded VLMs for Rigid-Body Motion Forecasting

Qi Sun and Jing Liao

Abstract—World models aim to predict how a scene evolves over time. Many video world models generate future frames, which is computationally expensive and does not explicitly represent object dynamics. In this paper, we instead study rigid-body motion forecasting that directly predicts multi-object trajectories. Existing learning-based simulators, however, acquire physical knowledge primarily from task-specific training data, which can limit generalization to unseen scenarios. To address this limitation, we introduce a trajectory-grounded vision-language model (VLM) that combines broad physical priors from foundation-model pretraining with trajectory supervision for accurate multi-object dynamics prediction. Our framework decomposes prediction into three levels: reason, plan, and animate. The VLM first reasons about object-centric physical events and predicts a sparse motion plan, after which a shape-aware flow-matching decoder animates the objects by generating dense per-object SE(3) trajectories. We further introduce RIGID+, a benchmark spanning eight categories of rigid-body phenomena, and evaluate generalization to unseen shapes, object counts, initial velocities, and compositions of physical phenomena. Experiments on MOVi and RIGID+ show that physical event reasoning and motion-plan conditioning improve trajectory prediction, with robust gains over learning-based simulators under distribution shift. Project page available at: <https://qisun.github.io/traj-vlm>

Index Terms—World models, rigid-body motion forecasting, vision-language models, flow matching.

I. INTRODUCTION

PREDICTING how objects move and interact is a central component of world models [1], with applications in robotics [2], [3], autonomous driving [4], and visual intelligence [5]–[7]. Current video world models commonly predict future frames with diffusion models [8]–[11]. Despite their strong visual fidelity, pixel-space models are computationally expensive and provide limited interpretability and control over object dynamics. Learning-based simulators offer an appealing alternative: they amortize simulation over data and predict dynamics directly from object states or geometry [12]–[15]. Unlike pixel-space video predictors, they produce explicit, view-independent motion that can be inspected, edited, and reused by downstream models.

Recent methods such as PhysCtrl [16] and PhysiFormer [17] predict complete geometric trajectories rather than rolling out one state at a time. However, their training and evaluation remain concentrated on narrow domains, such as single-object MPM dynamics [18] or free-fall and collision scenes in Kubric [19]. Because these models acquire physical knowledge primarily from task-specific simulation trajectories, it remains unclear whether they can transfer to diverse rigid-body

phenomena, compositions of familiar interactions, or shifts in geometry and initial conditions.

Foundation vision-language models (VLMs) [20], [21] provide a complementary source of prior knowledge. Internet-scale pretraining gives them broad object semantics, qualitative physical priors, and compositional reasoning ability. VLM-based motion planners can therefore infer plausible motion from images and language outside a fixed simulation domain [22]–[24]. However, foundation VLMs are trained primarily to predict language, not metrically calibrated state transitions. They may infer that an object will fall, collide, and settle, but cannot reliably determine the corresponding timing, translation, rotation, or contact response. Thus, the semantic breadth of a foundation VLM does not by itself provide temporally dense and numerically accurate motion forecasts.

We therefore introduce a *trajectory-grounded VLM* that is post-trained on object trajectories to produce motion plans from visual observations and optional physical conditions. To bridge qualitative physical reasoning and dense numerical motion forecasting, we organize prediction into three levels: reason, plan, and animate. First, waypoint supervision specifies where objects move but does not explicitly label the associated interactions; we therefore train the VLM to predict an object-centric physical event trace that summarizes their order. Second, event descriptions remain qualitative and do not uniquely determine metric motion; the VLM therefore predicts sparse 3D translation waypoints that form an explicit motion plan. We keep this plan sparse because predicting dense motion with an autoregressive VLM would require long sequences of tightly coupled coordinates, making generation expensive and prone to autoregressive error accumulation. Finally, the motion plan constrains low-frequency motion while leaving rotation and fine-grained contact dynamics unspecified. This plan reduces global trajectory ambiguity, allowing a lightweight shape-aware flow-matching decoder to model dense per-object SE(3) trajectories conditioned on coarse motion (Figure 1).

We evaluate our framework on MOVi and introduce RIGID+, a benchmark covering eight rigid-body phenomenon categories, together with an evaluation suite for shifts in shape, object count, initial velocity, and scene composition. Experiments demonstrate the benefits of physical event reasoning and motion-plan conditioning, particularly under distribution shift. We summarize our contributions as follows:

- We introduce a two-stage framework for rigid-body motion forecasting: a trajectory-grounded VLM reasons about physical events and plans sparse 3D motion, and a conditional decoder generates dense SE(3) trajectories. The explicit plan connects qualitative reasoning with numerical motion forecasting.

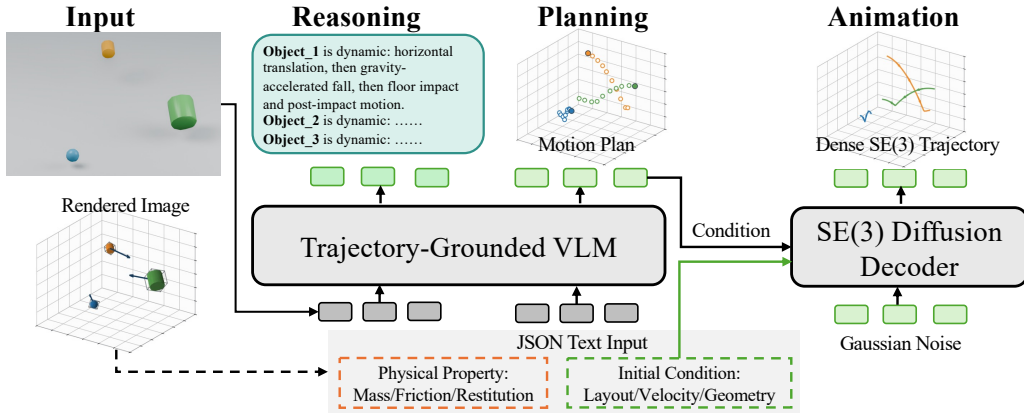


Fig. 1. **Overview of our two-stage reason, plan, and animate framework.** The trajectory-grounded VLM receives rendered observations, an object layout, and available physical conditions, including initial velocities, mass, friction, restitution, and external interventions. It first predicts an object-centric physical event trace and then a sparse 3D translation plan. The shape-aware flow-matching decoder combines this plan with object geometry, initial poses and scales, and initial linear and angular velocities to generate dense per-object SE(3) trajectories. Mass, friction, restitution, and interventions condition the VLM, not the decoder directly; their influence on dense motion is mediated by the predicted plan.

- We introduce object-centric physical-event supervision for motion planning. Controlled ablations show that predicting an event trace before waypoints improves translation accuracy over waypoint-only prediction, with consistent gains for free-text and structured event formats.
- We develop a shape-aware flow-matching decoder that jointly models object interactions and temporal dynamics conditioned on the motion plan. The complete framework achieves lower average translation and rotation errors than the evaluated learning-based baseline on MOVi, RIGID+, and the OOD suite; ablations support both explicit plan conditioning and geometry features.

II. RELATED WORK

A. Learning-based Physical Simulation.

Learning-based physical simulators learn state transitions or complete trajectories through graph, mesh, implicit-geometry, and particle representations, with related generative models operating directly on rigid transformations [13]–[15], [25]–[31]. Early work established the core relational formulation: Interaction Networks [32] and the Neural Physics Engine [33] introduced object- and relation-centric inductive biases for compositional dynamics. DPI-Net [34] then extended particle-based learning across rigid bodies, deformable objects, and fluids, while GNS [12] scaled graph-based autoregressive simulation across diverse physical systems. Recent work has shifted toward more scalable and unified trajectory models. RigidFormer [35] combines object-level tokens with differentiable rigid projection for efficient mesh-free rollout, while Unified Simulation [36] models multiple classes of Lagrangian dynamics with a shared Transformer. Generative formulations further predict entire trajectories: PhysCtrl [16] produces MPM particle trajectories as controls for video generation, and the concurrent PhysiFormer [17] generates world-coordinate vertex trajectories for rigid and elastic objects with a flow-matching model. In contrast to these direct low-level trajectory predictors, our hierarchy uses an explicit object-centric motion plan to connect transferable physical reasoning with dense SE(3) realization.

B. VLM-based Motion Reasoning and Planning.

Large vision-language foundation models (VLMs), such as Gemini [20], exhibit broad visual understanding and reasoning capabilities acquired from large-scale pretraining. Explicit intermediate reasoning further improves complex problem solving in large reasoning models and has been extended from multimodal mathematical reasoning to visual generation [37]–[42]. Building on these capabilities, recent work applies VLMs to motion prediction and control. VLIPP [22] asks GPT-4o to infer a coarse physically plausible motion plan and converts the plan into optical-flow guidance for video generation. WISA [43] decomposes physical knowledge into textual descriptions, qualitative categories, and quantitative attributes for physics-aware video generation. MotiMotion [23] uses visual reasoning to infer motion controls for controllable video generation. MolmoMotion [44] forecasts 3D point trajectories from visual observations and language instructions. LAMP [24] translates language instructions into explicit motion plans for controllable video synthesis. WorldDirector [45] uses an LLM to orchestrate continuous 3D object and camera trajectories as spatial conditions for a video world model. These approaches primarily use VLM priors for visual motion control or generation rather than physically evaluated object-state forecasting. In contrast, we introduce trajectory-grounded VLMs that learn structured interaction events and object-centric motion plans as an explicit reasoning interface for dense rigid-body motion forecasting.

III. METHOD

Our method uses two learned stages to implement three prediction levels: reason, plan, and animate. A trajectory-grounded VLM first *reasons* about object-centric physical events to produce an event trace and then *plans* sparse motion through 3D translation waypoints. A shape-aware flow-matching decoder finally *animates* the objects by generating dense SE(3) trajectories. The motion plan provides an explicit interface between event-level reasoning and dense trajectory generation.

A. Problem Formulation

We study rigid-body motion forecasting: given a scene with M rigid objects, we predict their future motion over H frames. The observed scene $\mathcal{S}$ comprises rendered observations, initial object poses, and known object geometry and scale. Optional physical conditions $\mathcal{A}$ include initial linear and angular velocities, mass, friction, restitution, and external interventions when available. The VLM receives these conditions as text. We denote by $\mathcal{D}$ the decoder’s direct scene inputs: object geometry, initial poses and scales, and initial linear and angular velocities. Other physical conditions reach the decoder only through the predicted motion plan. We represent the dense trajectory as

$$\mathbf{X} \in \mathbb{R}^{M \times H \times 9}, \quad (1)$$

where each object-frame state concatenates a 3D translation and a continuous 6D rotation representation. This 9D Euclidean encoding represents the trajectory of rigid poses in SE(3).

The task assumes access to object geometry and initial layout, but not to a complete simulator state. When quantities such as mass, friction, restitution, or applied forces are unavailable, prediction is conditioned on the observed scene and the supplied subset of physical conditions. The goal is to generate plausible dynamics under this partial specification, rather than to solve a fully specified initial-value problem.

Directly generating H dense poses with a VLM is inefficient and requires the language model to represent small numerical changes. We instead introduce a sparse motion plan

$$\mathbf{P} \in \mathbb{R}^{M \times K \times 3}, \quad K \ll H, \quad (2)$$

where the K waypoints are defined at fixed future times in the global coordinate system. The planner’s waypoint targets are obtained by temporally subsampling the translations of the ground-truth trajectory. We denote the object-centric physical event trace by $\mathbf{R}$ and factorize the joint prediction as

$$p(\mathbf{X}, \mathbf{P}, \mathbf{R} \mid \mathcal{S}, \mathcal{A}) = p_\theta(\mathbf{X} \mid \mathbf{P}, \mathcal{D}) p_\phi(\mathbf{P} \mid \mathbf{R}, \mathcal{S}, \mathcal{A}) p_\phi(\mathbf{R} \mid \mathcal{S}, \mathcal{A}), \quad (3)$$

Here, p_ϕ models both event and waypoint generation, and p_θ is the dense trajectory decoder. The decoder receives event information through the numerical plan $\mathbf{P}$, rather than the textual trace $\mathbf{R}$. This factorization is a modeling assumption: once the plan and $\mathcal{D}$ are given, the decoder does not separately observe the remaining physical conditions. This division assigns interaction sequences and coarse translations to the VLM, while the decoder generates dense translations and rotations conditioned on the plan and initial object state.

B. Trajectory-Grounded Reasoning and Planning

We use a pretrained VLM to model the first two levels in Equation 3. Supervised fine-tuning on object trajectories grounds the VLM to predict a physical event trace $\mathbf{R}$ followed by the motion plan $\mathbf{P}$. The VLM does not predict dense poses or mesh vertices.

1) *Visual scene conditioning*: The planner receives rendered observations as visual input. The object layout and optional conditions $\mathcal{A}$ are serialized as text and provided in the same prompt. This input retains appearance and shape cues that are absent from a state-only description, while the explicit layout and conditions specify the coordinate frame and known physical state. We keep the input format fixed across different physical phenomena.

2) *Physical event reasoning*: Waypoint-only supervision specifies where an object moves, but does not explicitly describe the physical events associated with its motion. We therefore train the VLM to first produce an object-centric sequence of physical events. The trace describes motion states, external interventions, interactions, and responses, such as release, collision, sliding, and settling; the full ontology is listed in the supplementary appendix. Its ordering provides an explicit summary of the interaction sequence for subsequent waypoint prediction through $p_\phi(\mathbf{P} \mid \mathbf{R}, \mathcal{S}, \mathcal{A})$.

Training traces are derived from simulator metadata and ground-truth trajectories using the procedure in Section IV-B. They retain qualitative event types and categorical attributes but exclude future numerical states and event timestamps. These traces are supervision targets, not test-time inputs.

We serialize the same ordered events either as a per-object JSON record (*structured*) or as a natural-language sentence (*free-text*). Both formats share the same ontology and event content, differing only in serialization. We use free-text reasoning in the final model.

3) *Motion plan prediction*: For object i , the planner predicts $\mathbf{P}_i \in \mathbb{R}^{K \times 3}$. We use $K = 9$ waypoints by default. Each waypoint is a global 3D translation at a fixed future time; rotation is left to the dense decoder. We first normalize the translation coordinates and then discretize each normalized coordinate with a fixed 256-bin scalar quantizer. We denote this normalization-and-quantization mapping by Q and generate the resulting bin indices with tokens from the VLM’s original vocabulary. This follows the general principle of keeping control targets aligned with the pretrained VLM output space [46], and avoids introducing a separate action-token vocabulary.

During supervised fine-tuning, the target sequence contains the physical event trace followed by the quantized waypoints,

$$\mathbf{y} = \text{Serialize}(\mathbf{R}, Q(\mathbf{P})). \quad (4)$$

We train the planner with the standard autoregressive token loss

$$\mathcal{L}_{\text{plan}} = - \sum_{j=1}^{|\mathbf{y}|} \log p_\phi(y_j \mid y_{<j}, \mathcal{S}, \mathcal{A}). \quad (5)$$

At inference, the planner generates the event trace and quantized waypoints autoregressively from the observed scene and available physical conditions. Each generated bin index is dequantized to the left endpoint of its bin in normalized coordinate space. We then apply inverse normalization to recover the global 3D coordinates of the motion plan supplied to the dense decoder.

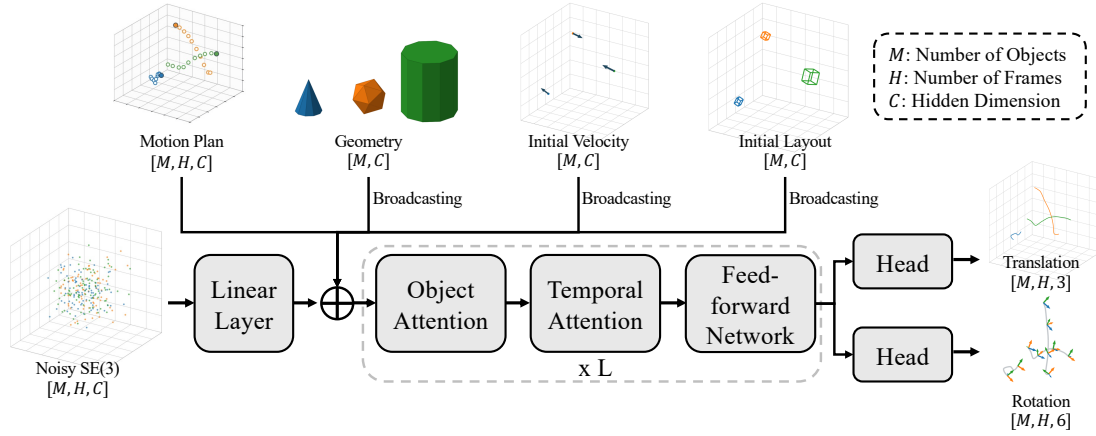


Fig. 2. **Architecture of the shape-aware flow-matching decoder.** The diffusion transformer (DiT) predicts clean, dense object trajectories from noisy poses, conditioned on the interpolated motion plan, reference geometry, initial velocities, and layout. Conditions are embedded into C -dimensional features and added to the noisy-pose tokens, with object-level features broadcast across frames. Each of the L blocks applies object attention, temporal attention, and a feed-forward network. Separate heads output translations in $\mathbb{R}^{M \times H \times 3}$ and 6D rotation representations in $\mathbb{R}^{M \times H \times 6}$, where M and H denote the numbers of objects and frames, respectively. Diffusion-time adaLN modulation is omitted from the diagram for clarity.

C. Shape-Aware Flow-Matching Decoder

The sparse motion plan represents each object’s coarse displacement and can constrain interaction timing, but it does not explicitly encode event labels, dense motion, rotation, or contact-scale corrections. We therefore use the shape-aware conditional flow-matching decoder in Figure 2 to map the plan and initial object state to the dense trajectory $\mathbf{X}$. The decoder operates jointly over all M objects and H time steps, allowing it to model both inter-object coupling and temporal evolution.

1) *Explicit motion-plan conditioning:* We linearly interpolate the K translation waypoints to the dense horizon,

$$\tilde{\mathbf{P}} = \text{Interp}_H(\mathbf{P}) \in \mathbb{R}^{M \times H \times 3}. \quad (6)$$

Condition-specific multilayer perceptrons (MLPs) project the heterogeneous inputs into a shared C -dimensional feature space. A motion-plan MLP maps each interpolated 3D position in $\tilde{\mathbf{P}}$ to a feature vector, yielding $\mathbf{E}_P \in \mathbb{R}^{M \times H \times C}$. For geometry, we first extract a feature from each reference mesh using a shape encoder [47] and then project it with a separate MLP. The raw initial linear and angular velocities and initial layout (pose and scale) are encoded by their respective MLPs. These object-level features are broadcast over the H frames and summed with $\mathbf{E}_P$ and a linear embedding of the noisy poses. The resulting tensor has shape $M \times H \times C$, with one condition-enriched token per object-frame pair, as illustrated in Figure 2. We denote the collection of these decoder conditions, obtained from $(\mathbf{P}, \mathcal{D})$, by $\mathbf{C}$.

This explicit numerical interface differs from passing VLM hidden states or a KV cache to a low-level decoder [48], [49]. Following the conditioning principle of T2I-Adapter [50] and ControlNet [51], it keeps the planned trajectory interpretable and independently editable. The plan serves as a condition, not a base trajectory for residual prediction. Given a noisy trajectory $\mathbf{Z}_\tau$ and flow time τ , the decoder predicts the complete clean pose trajectory

$$\hat{\mathbf{X}}_\theta(\mathbf{Z}_\tau, \tau, \mathbf{C}) = [\hat{\mathbf{T}}_\theta, \hat{\mathbf{R}}_\theta], \quad (7)$$

where $\hat{\mathbf{T}}_\theta \in \mathbb{R}^{M \times H \times 3}$ and $\hat{\mathbf{R}}_\theta \in \mathbb{R}^{M \times H \times 6}$ are the predicted translation and continuous 6D rotation representations, respectively.

2) *Shape-aware diffusion transformer:* Objects with the same coarse translation plan can respond differently because of their geometry, scale, pose, and velocity. The DiT backbone processes the condition-enriched tokens through L repeated blocks, each consisting of object attention, temporal attention, and a feed-forward network (FFN), as shown in Figure 2. Object attention performs self-attention across the M objects at each fixed frame, allowing each object’s representation to incorporate the geometry and motion of other objects and model interaction-dependent dynamics such as collisions. Temporal attention then performs self-attention across the H frames of each object, relating its poses before and after an interaction; temporal RoPE [52] encodes frame order. Alternating these two attention axes propagates interaction information across objects and time while preserving the $M \times H \times C$ token layout. The FFN applies a shared nonlinear transformation independently to each object-frame token, mixing feature channels to update the representation without exchanging information across objects or frames. The flow time τ , distinct from the frame index, is encoded with a sinusoidal embedding followed by an MLP and injected into every block through adaptive layer normalization (adaLN), modulating all tokens according to the current noise level. Finally, separate linear heads map each token to its 3D translation and 6D rotation representation in Equation 7.

3) *Flow-matching training and inference:* We train the decoder by conditional flow matching in the Euclidean pose-encoding space. Let $\mathbf{X}$ be a clean trajectory and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ be Gaussian noise of the same shape. For $\tau \in [0, 1)$, we define the linear probability path and its target velocity as

$$\mathbf{Z}_\tau = \tau \mathbf{X} + (1 - \tau) \epsilon, \quad \mathbf{u}_\tau = \frac{\partial \mathbf{Z}_\tau}{\partial \tau} = \mathbf{X} - \epsilon. \quad (8)$$

Following JiT [53], the network is parameterized to predict the clean trajectory rather than the velocity directly. The

corresponding velocity field is

$$\mathbf{v}_\theta = \frac{\hat{\mathbf{X}}_\theta(\mathbf{Z}_\tau, \tau, \mathbf{C}) - \mathbf{Z}_\tau}{1 - \tau}, \quad (9)$$

and we minimize the flow-matching objective

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\mathbf{X}, \epsilon, \tau} \left[\|\mathbf{v}_\theta - \mathbf{u}_\tau\|_2^2 \right]. \quad (10)$$

At inference, we initialize $\mathbf{Z}_0$ with Gaussian noise and integrate $d\mathbf{Z}_\tau/d\tau = \mathbf{v}_\theta$ from $\tau = 0$ to 1 using 25 explicit Euler steps on a uniform time grid with step size $1/25$. The velocity field is evaluated at $\tau = 0, 1/25, \dots, 24/25$, yielding 25 function evaluations without evaluating the velocity field at $\tau = 1$. The final state $\mathbf{Z}_1$ yields the dense translation and rotation trajectories. Network sizes and optimization settings are provided in Section V-A.

IV. DATASET CURATION

A. Rigid-body benchmark

Previous methods are primarily evaluated on Kubric [19], which emphasizes free fall and simple object collisions and therefore provides limited coverage of rigid-body dynamics. To support a more comprehensive evaluation, we curate RIGID+, a PyBullet [54]-based dataset covering eight categories of physical phenomena: *free motion, impact and collision, sustained contact and friction, stability and support, external intervention, constrained rigid-body motion, many-body interaction, and boundary interaction*. Together, these scenes capture ballistic motion, momentum transfer, sliding and rolling, stacking and toppling, force-driven motion, articulated constraints, collision propagation, and repeated environmental contacts. Further details of the scene construction and physical parameter ranges are provided in the appendix.

B. Reasoning annotation

We automatically derive event-reasoning supervision from simulator ground truth using a shared ontology of 28 physical-process primitives. The ontology describes four complementary aspects of rigid-body dynamics: motion states (e.g., *ballistic motion, sliding, rolling*), external actions and constraints (e.g., *force application, support removal, constrained motion*), interactions (e.g., *object collision, boundary collision, support contact*), and physical responses (e.g., *momentum transfer, rebound, stability failure*). For each dynamic object, we combine scripted intervention and constraint metadata, ground-truth contact episodes, and state changes inferred from its position, orientation, and velocity trajectories. The detected primitives are assigned to the corresponding object and ordered by onset time to form an object-centric physical event trace. Multiple primitives may describe the same transition when complementary processes co-occur, such as an object collision followed by momentum transfer and a direction change. Labels are generated deterministically for both MOVi and RIGID+, with consistency checks on simulator replay, object alignment, temporal ordering, and ontology validity. We retain only qualitative primitive types and categorical attributes, excluding future coordinates, velocities, forces, contact locations, and

TABLE I
COMPARISON ON MOVi. VALUES ARE TRANSLATION RMSE ($\times 10^{-2}$ M); THE LEARNING-BASED SIMULATOR RESULTS ADDITIONALLY REPORT ROTATION RMSE (DEGREES) AFTER "/. LOWER IS BETTER.

Method	MOVi-A	MOVi-B	MOVi-S	Avg.
<i>VLM-based Planner</i>				
VLIPP (GPT-4o)	132.5	119.4	139.1	130.3
WorldDirector (GPT-5.5)	127.9	30.49	94.1	84.16
WorldDirector (Gemini 3)	129.1	124.8	138	130.6
Ours (VLM)	14.36	25.85	24.38	21.53
<i>Learning-based Simulator</i>				
PhysiFormer	5.692 / 8.388	7.133 / 11.48	5.366 / 7.198	6.064 / 9.022
Ours	4.356 / 6.743	6.759 / 6.380	4.013 / 5.64	5.043 / 6.254

event timestamps to prevent direct trajectory leakage. These labels serve as privileged training targets and are not available at test time.

V. EXPERIMENTS

A. Implementation Details

1) *Model Architecture Details*: We use Qwen3.5-2B [55], a VLM with reasoning capability, as the planner backbone without changing the network architecture.

The shape-aware flow-matching decoder is a 6-layer Transformer with 8 attention heads, a hidden dimension of 768, and an MLP expansion ratio of 4. The flow time is encoded using a 256-dimensional sinusoidal embedding followed by an MLP and injected into every Transformer block through adaLN modulation. The motion-plan length is $K = 9$, while the full trajectory length is $H = 49$.

2) *Training Details*: We finetune the VLM for 5 epochs using LoRA [56] with rank 16, scaling factor 32, and dropout 0.05 in its attention layers. We keep its vision encoder frozen during finetuning. During VLM finetuning, we use AdamW [57] with a learning rate of 10^{-5} , a global batch size of 24, and $8 \times$ NVIDIA A6000 GPUs. We train the flow-matching decoder for 25,000 iterations with a batch size of 256 and select the checkpoint with the lowest validation loss. During training, we use log-normal noise-level sampling following the noise-scheduling approach of SD3 [58]. At inference, we use explicit Euler integration with 25 function evaluations (NFEs) on a uniform time grid by default.

The training/inference code, full training data, and pre-trained models would be available upon publication.

B. Experimental Setup

1) *Benchmarks and Metrics*: Following prior work on learning-based rigid-body simulation [17], [35], we evaluate models on the MOVi-A/B/S benchmarks. Each benchmark consists of 125 test cases. We also evaluate on RIGID+, which contains 800 test cases across 8 categories of rigid-body phenomena. We report the root mean squared error (rMSE) of translation and rotation, with rotation error measured in degrees, between predicted and ground-truth trajectories. Translation rMSE is reported in units of 10^{-2} m; for example, a table value of 3.381 corresponds to 0.03381 m. The VLM-planner rows report translation only, whereas the complete

TABLE II

COMPARISON ON THE RIGID+ TEST SPLIT. VALUES ARE TRANSLATION RMSE ($\times 10^{-2}$ M); THE LEARNING-BASED SIMULATOR RESULTS ADDITIONALLY REPORT ROTATION RMSE (DEGREES) AFTER “/”. LOWER IS BETTER.

Method	Boundary	Constrained	External	Free	Impact	Many-body	Stability	Contact	Avg.
<i>VLM-based Planner</i>									
VLIPP (GPT-4o)	197.9	282.2	252.7	197.7	152.4	185.2	162.3	201.3	204.0
WorldDirector (GPT-5.5)	113.6	55.35	95.5	39.79	94.92	123.7	120.8	136.1	97.47
WorldDirector (Gemini 3)	152.1	179.1	133.4	169.1	135.4	122.6	164.3	119.7	147.0
Ours (VLM)	26.44	5.15	5.805	30.79	11.59	22.08	6.835	10.13	14.85
<i>Learning-based Simulator</i>									
PhysiFormer	12.55 / 8.79	3.199 / 13.82	2.290 / 11.05	14.57 / 7.734	8.040 / 12.85	6.649 / 9.411	3.033 / 7.578	3.137 / 14.41	6.684 / 10.71
Ours	8.004 / 8.622	3.120 / 1.546	1.836 / 7.357	1.798 / 8.937	3.941 / 11.34	2.991 / 8.502	2.193 / 0.665	3.165 / 7.751	3.381 / 6.84

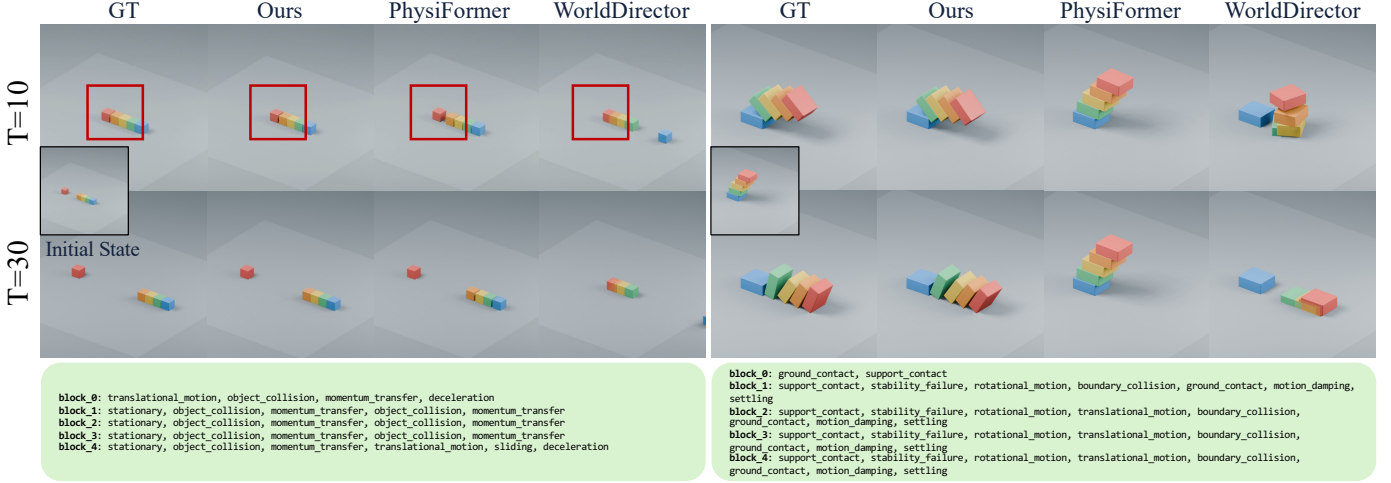


Fig. 3. **Qualitative comparison of collision propagation and stack instability on RIGID+.** The left scene shows motion propagating through a row of objects after impact; the right scene shows a stack losing stability. Our predictions capture these interaction-dependent changes in motion. In these examples, PhysiFormer moves downstream objects without the initiating collision and misses stack instability, while zero-shot VLM planners exhibit object interpenetration.

simulators report both translation and rotation. The “Avg.” columns summarize the listed benchmarks, categories, or OOD axes with an unweighted arithmetic mean; the OOD average therefore weights each axis equally despite different test-set sizes.

2) *Baselines:* We compare with both VLM planners and learning-based simulators to evaluate prediction accuracy and generalization. PhysiFormer [17] is a concurrent learning-based simulator that predicts dense vertex trajectories for rigid and elastic objects conditioned on their initial states. For zero-shot planning, we evaluate VLIPP [22] with GPT-4o and WorldDirector [45]¹ with GPT-5.5 and Gemini 3.0. Given the physical state at frame 0, including object geometry, pose, linear and angular velocities, each VLM predicts object positions and orientations at 9 future anchor frames. We linearly interpolate positions and use SLERP [59] for orientations to obtain the full 49-frame trajectory.

We compare against PhysiFormer because it is the closest publicly available full-trajectory simulator with a compatible rigid-body prediction setting. We reimplement its training pipeline² using the official codebase and train it on our data splits. We do not include RigidFormer because its official implementation is not publicly available; other cited simulators either lack public implementations or use substantially different particle-, mesh-, or rollout-based task interfaces that

do not support a controlled comparison without extensive reimplementations.

C. Rigid-Body Motion Forecasting

Tables I and II compare the VLM planner and the complete model on MOVi and RIGID+, respectively. Trajectory grounding substantially improves the VLM planner over zero-shot baselines, while our complete model reduces the average translation and rotation errors of PhysiFormer on both benchmarks. These gains are not uniform across every metric: on RIGID+, PhysiFormer has slightly lower translation error for sustained contact and lower rotation error for free motion. Figure 3 illustrates the distinction between coarse motion plausibility and interaction-consistent trajectories. In the collision example, PhysiFormer moves downstream objects without the initiating contact; in the stack example, it misses the loss of stability. Our model captures both transitions, while the zero-shot planners exhibit interpenetration. The proposed hierarchy provides a possible explanation: event supervision encourages the planner to represent interaction order, the numerical plan constrains global displacement, and joint object-time decoding uses geometry to model coupled translation and rotation. This interpretation is supported by the event-reasoning and decoder ablations in Tables V and VI, rather than by the qualitative examples alone. It does not imply that the model enforces contact constraints or recovers a causal physical

¹<https://github.com/pPetrichor/WorldDirector>

²<https://github.com/yimingc9/PhysiFormer/tree/main>

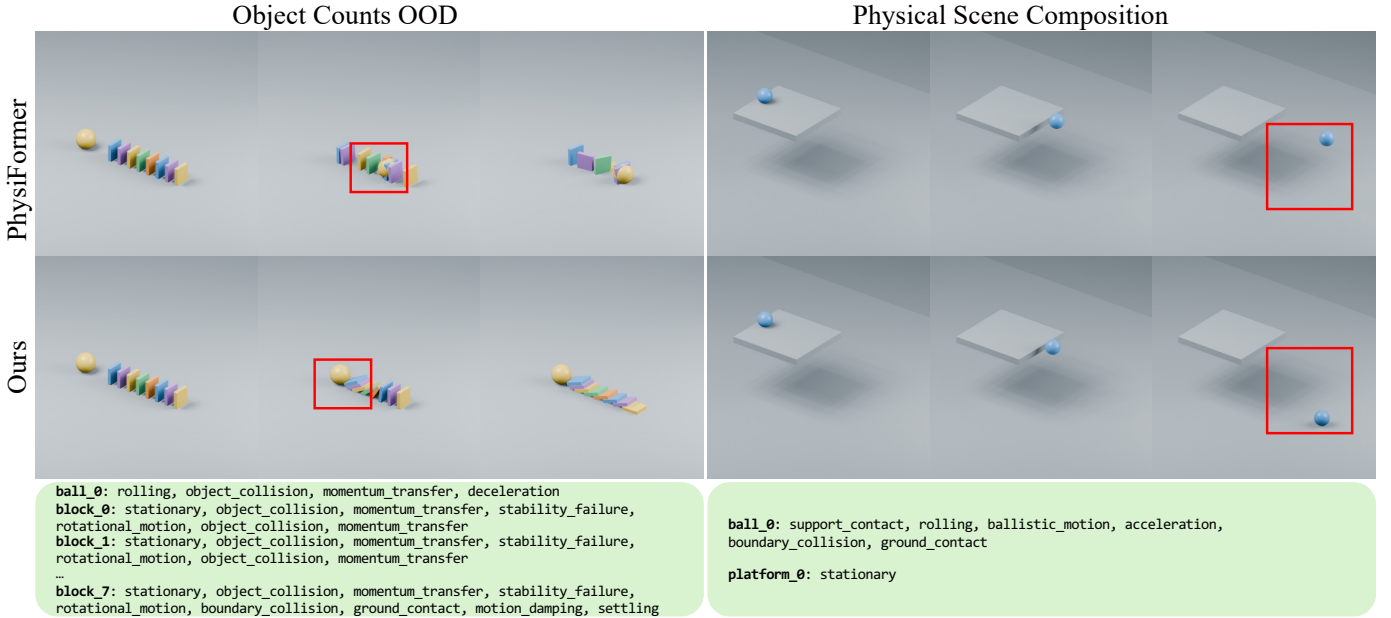


Fig. 4. **Qualitative comparison under out-of-distribution object counts and scene compositions.** The left example tests collision propagation along a longer row of upright blocks; the right tests motion after an object loses contact with an elevated platform. Red boxes highlight the contrasting outcomes: successive block rotations versus interpenetration (left), and falling motion versus approximately constant-velocity extrapolation after contact ends (right). The event sequences below the images describe collision-driven momentum transfer and stability failure (left), and the transition from support contact to ballistic motion (right).

TABLE III

OUT-OF-DISTRIBUTION RESULTS. VALUES ARE TRANSLATION RMSE ($\times 10^{-2}\text{M}$); THE LEARNING-BASED SIMULATOR RESULTS ADDITIONALLY REPORT ROTATION RMSE (DEGREES) AFTER "/>. LOWER IS BETTER.

Method	Novel Shapes	Initial Velocities	Object Counts	Scene Composition	Avg.
WorldDirector [45] (Gemini 3.0)	205.3	132.4	66.19	66.19	117.5
WorldDirector [45] (GPT-5.5)	192.8	141.0	70.39	136.7	135.2
Ours (VLM)	13.62	35.93	8.250	41.82	24.91
PhysiFormer [17]	15.69 / 38.88	23.33 / 57.75	5.326 / 30.99	65.41 / 87.90	27.44 / 53.88
Ours	10.62 / 26.19	18.01 / 11.94	5.489 / 16.91	17.66 / 29.61	12.94 / 21.16

law. Additional qualitative comparisons are provided in the supplementary appendix.

D. Out-of-distribution Generalization

Table III evaluates generalization to novel shapes, initial velocities, object counts, and compositions of physical phenomena. Trajectory grounding improves the VLM planner across all settings, and our complete model outperforms PhysiFormer except in translation under novel object counts. In Figure 4, our model captures successive domino rotations and the transition to falling after support contact ends, whereas PhysiFormer produces interpenetration or approximately constant-velocity extrapolation in these examples. The event sequences beneath the images make the relevant transitions explicit, linking collision and momentum transfer to toppling, and loss of support to ballistic motion. These examples illustrate the intended role of event-conditioned planning, but do not establish accurate event prediction across the full OOD suite.

E. Parameter Sensitivity

Our framework responds to changes in the specified physical parameters, producing distinct motion outcomes rather than a fixed response for a given scene. Mass, friction, and

restitution are supplied to the VLM as text; their effect on the decoder is mediated by the resulting motion plan, not by direct parameter inputs. Figure 5 illustrates this sensitivity to object mass, friction, and restitution. Changing the colliding spheres from equal masses (1 kg each) to unequal masses (4 kg and 1 kg) alters their post-impact motion, with the lighter sphere traveling farther in the unequal-mass case. Increasing the friction coefficient from 0.1 to 0.9 reduces the sliding distance of the cube, while increasing the restitution coefficient from 0.1 to 0.9 produces a more pronounced rebound of the falling sphere. These examples demonstrate qualitative sensitivity to physical-parameter conditions, rather than quantitative calibration across the full parameter range.

F. Ablation Studies

1) *Sparse motion plan:* Table IV compares uniform temporal downsampling at different intervals d . Increasing d shortens the output sequence, but an overly sparse motion plan under-constrains the motion between supervised waypoints: $d = 25$ attains the lowest supervised-point rMSE but the highest full-trajectory rMSE. Point-level errors are evaluated at different temporal subsets and are therefore not directly comparable across d ; full-trajectory rMSE measures the common prediction task. The dense representation ($d = 1$) deviates from

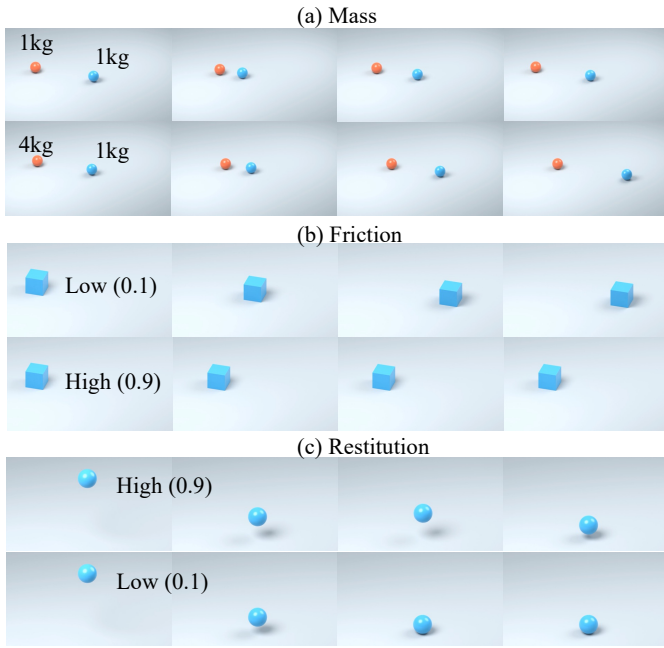


Fig. 5. **Sensitivity to physical parameters.** Columns show successive frames from left to right, with paired rows comparing different parameter settings. (a) Mass: two 1 kg spheres versus a 4 kg sphere colliding with a 1 kg sphere, yielding different post-impact motions. (b) Friction: the cube slides farther with a low friction coefficient (0.1) than with a high coefficient (0.9). (c) Restitution: the sphere exhibits a more pronounced rebound with a high restitution coefficient (0.9) than with a low coefficient (0.1). Our framework produces distinct motion outcomes in response to these parameter changes.

the ground-truth trajectory in its later portion, consistent with autoregressive error accumulation. Using $d = 5$ gives the lowest full-trajectory rMSE while reducing the average token count from 2147 to 453.7 relative to dense supervision ($d = 1$). Figure 6 qualitatively shows the same trade-off: very sparse representations omit important changes in motion, whereas our nine-waypoint representation retains the overall trajectory structure with substantially fewer points. We therefore use nine sparse waypoints ($d = 5$) as the default representation considering both efficiency and trajectory-level accuracy. We also compare downsampled waypoints with FAST [60], a trajectory tokenization method based on the discrete cosine transform (DCT) and byte-pair encoding (BPE). FAST uses the fewest output tokens (105.5) but has higher full-trajectory rMSE than our default representation (17.47 versus 15.95). Thus, the selected waypoint representation trades a longer output sequence for better trajectory accuracy in this comparison; the experiment does not isolate the effect of vocabulary changes.

2) *Physical event reasoning*: We compare the free-text and structured JSON formats during supervised fine-tuning to evaluate whether intermediate event reasoning improves downstream trajectory prediction. As shown in Table V, the two formats perform comparably: their MOVi average rMSEs are 16.49 and 16.61, respectively, and both substantially outperform direct waypoint prediction without reasoning (40.34). The same conclusion holds on RIGID+, where free-text and structured reasoning obtain 10.78 and 11.63, compared with 32.20 without reasoning. These results support our claim that event reasoning improves downstream trajectory prediction;

TABLE IV
ABLATION OF THE SPARSE MOTION PLAN PREDICTED BY THE VLM PLANNER. WE REPORT THE NUMBER OF SUPERVISED POINTS, TRANSLATION RMSE ($\times 10^{-2}$ M) AT THE SUPERVISED WAYPOINTS AND OVER THE FULL TRAJECTORY, AND THE NUMBER OF OUTPUT TOKENS. POINT RMSE USES DIFFERENT TEMPORAL SUBSETS ACROSS INTERVALS d ; TRAJECTORY RMSE EVALUATES THE COMMON DENSE HORIZON. LOWER IS BETTER FOR RMSE AND TOKEN COUNT.

Representation	#Points	Point rMSE	Trajectory rMSE	Tokens
Downsample ($d = 1$)	48	24.9	24.9	2147
Downsample ($d = 2$)	24	16.79	16.14	1105
Downsample ($d = 5$, Ours)	9	15.92	15.95	453.7
Downsample ($d = 10$)	4	14.77	17.28	237.5
Downsample ($d = 25$)	1	11.38	42.69	106.6
FAST ($d = 1$)	48	17.47	17.47	105.5

we use the slightly better free-text format in the final model. We evaluate the event trace through its effect on trajectory prediction rather than claiming standalone event-recognition accuracy. The *Pretrained VLM* control uses the off-the-shelf model without trajectory-grounding post-training. Its higher errors support the benefit of task-specific grounding over generic pretraining alone. This comparison does not isolate the contribution of pretraining relative to learning the same architecture from scratch. To reduce the cost of this controlled study, the fine-tuned variants in Table V are trained for four epochs; the pretrained control is not fine-tuned. The main planner in Tables I and II is trained for five epochs, so cross-table differences should not be attributed to the ablated component alone. The qualitative examples in Figure 7 further show that removing physical-event reasoning or visual conditioning changes the predicted paths and interaction geometry, consistent with the higher translation errors of these variants.

3) *Design of VLM guidance*: We ablate how trajectory-grounded VLM guidance conditions the flow-matching decoder in Table VI. Removing the motion plan increases the mean translation and rotation errors from 3.38/6.84° to 53.81/30.34°, supporting the importance of plan conditioning in the evaluated setting. Replacing the motion plan with an implicit VLM feature also increases error. This result favors the tested explicit interface, but does not rule out stronger implicit conditioning designs. Predicting translation residuals around the interpolated motion plan improves translation in several categories, but gives higher mean translation and rotation errors than direct prediction. Removing geometry features increases average translation rMSE from 3.38 to 5.54 and rotation rMSE from 6.84° to 10.82°, supporting shape-aware conditioning beyond the coarse translation plan, although some individual categories improve without geometry. We therefore use the motion plan as an explicit condition and directly predict the dense trajectory. Figure 8 shows that this design also follows the GT paths and final poses more closely, whereas the alternative conditioning strategies produce larger trajectory or orientation deviations.

G. Applications: Physics-Guided Video Generation

Our explicit object trajectories provide a spatial interface for controllable video generation [61]. We animate the scene geometry along the predicted SE(3) trajectories and render a depth video, which serves as a spatial condition for MiniMax-H3 [62] alongside the first-frame image. This interface allows

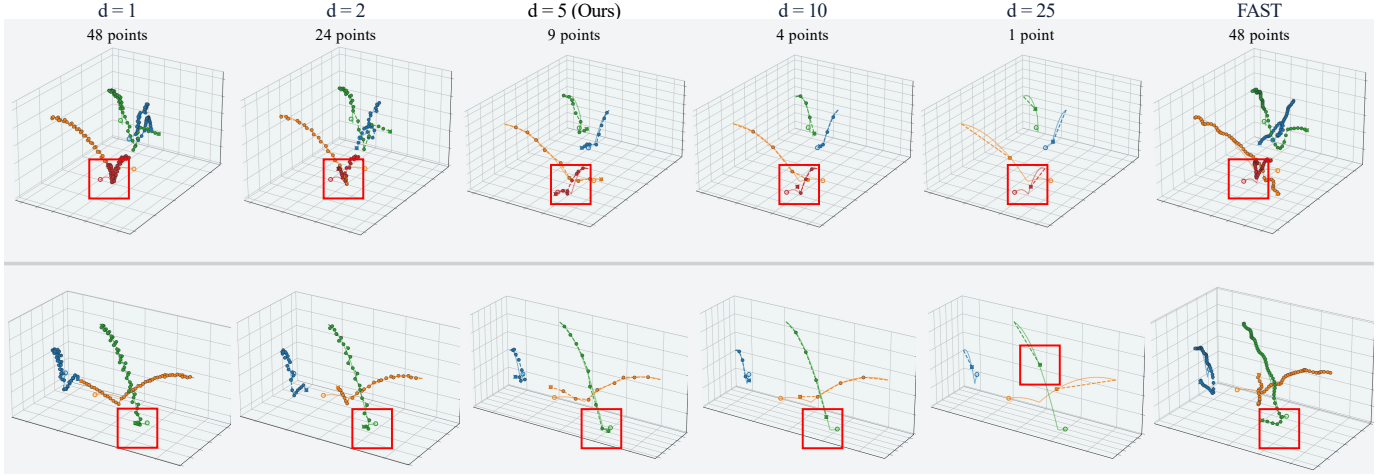


Fig. 6. **Ablation of the sparse motion-plan representation.** Each row shows one scene, comparing uniform temporal downsampling at interval d with FAST tokenization. **GT**: solid lines denote ground-truth trajectories. **Our prediction**: dots indicate motion-plan waypoints predicted by the VLM, and dashed lines denote their interpolated trajectories. Dense prediction ($d = 1$) shows larger late-trajectory deviations, while very sparse prediction ($d = 25$) misses changes in motion; FAST also produces larger trajectory deviations in these examples. Our default $d = 5$ representation preserves the overall motion with nine waypoints.

TABLE V

ABLATION OF THE TRAJECTORY-GROUNDED VLM PLANNER. VALUES ARE TRANSLATION RMSE ($\times 10^{-2}$ M); LOWER IS BETTER. FINE-TUNED VARIANTS USE FOUR EPOCHS, WHEREAS THE PRETRAINED CONTROL IS EVALUATED WITHOUT FINE-TUNING.

Model Variant	MOVi-A	MOVi-B	MOVi-S	MOVi Avg.	RIGID+ Avg.
Ours (Free-text Events)	8.61	26.25	14.61	16.49	10.78
Ours (JSON-structured Events)	8.31	26.82	14.70	16.61	11.63
Ours w/o Reasoning	40.02	50.54	30.46	40.34	32.20
Ours w/o Visual Conditioning	29.84	28.49	20.50	26.28	27.32
Pretrained VLM	70.51	135.32	96.60	100.81	77.45

TABLE VI

ABLATION OF MOTION-PLAN AND GEOMETRY CONDITIONING ON RIGID+. EACH CELL REPORTS TRANSLATION RMSE ($\times 10^{-2}$ M) / ROTATION RMSE (DEGREES). LOWER IS BETTER; THE BEST RESULT FOR EACH METRIC IS BOLD.

Variant	Boundary	Constrained	External	Free	Impact	Many-body	Stability	Contact	Avg.
Ours	8.004 / 8.622	3.129 / 1.546	1.836 / 7.357	1.798 / 8.937	3.941 / 11.34	2.991 / 8.502	2.193 / 0.665	3.165 / 7.751	3.381 / 6.84
Ours w/ Residual Prediction	10.96 / 15.11	3.044 / 1.086	9.368 / 5.964	11.99 / 10.54	3.692 / 5.364	2.934 / 8.108	7.236 / 5.436	8.024 / 8.708	7.156 / 7.539
Ours w/ Implicit Conditioning	16.41 / 20.35	33.78 / 34.87	70.04 / 53.58	14.25 / 60.24	10.33 / 24.52	77.57 / 56.54	14.91 / 68.6	19.34 / 43.49	32.08 / 45.27
w/o Motion Plan	74.7 / 23.61	31.81 / 23.51	18.67 / 30.15	121 / 57.05	45.44 / 10.38	31.98 / 19.83	83.3 / 38.62	23.57 / 39.59	53.81 / 30.34
w/o Geometry	9.9570 / 23.6800	8.6740 / 7.2280	2.4840 / 20.1300	7.7090 / 9.0320	4.7856 / 8.8740	3.5880 / 1.4950	3.2890 / 8.0750	3.8320 / 8.0758	5.5398 / 10.8237

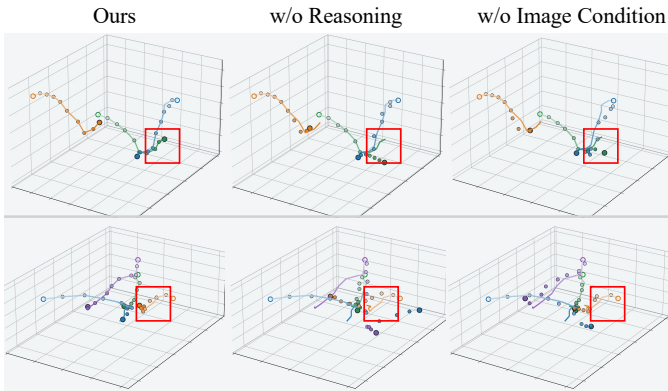


Fig. 7. **Ablation of physical-event reasoning and visual conditioning.** The two rows show different scenes, comparing the full model with variants that omit the event trace or the input image. **GT**: solid lines show ground-truth trajectories. **Our prediction**: dots indicate predicted motion-plan waypoints. Removing either input changes the predicted object paths and interaction geometry, whereas the full model more closely follows the GT motion.

the video generator to synthesize appearance while using our predicted dynamics to guide object motion.

Figure 9 compares video generation with and without this depth condition, using the same first frame for each scene. Without spatial conditioning, the four upright tablets become three in the left scene, while the four spheres become five in the right scene, disrupting the intended interaction sequence. With our trajectory guidance, the generated frames better preserve object persistence and collision-driven motion in both the toppling sequence (left) and the cargo-sphere interaction (right). These qualitative examples illustrate how explicit trajectory control can improve the physical plausibility of generated videos.

H. Runtime Analysis

Our autoregressive planner generates the physical event trace and motion plan, after which the flow-matching decoder predicts the dense trajectory. This semantic reasoning and planning stage introduces additional latency relative to a direct

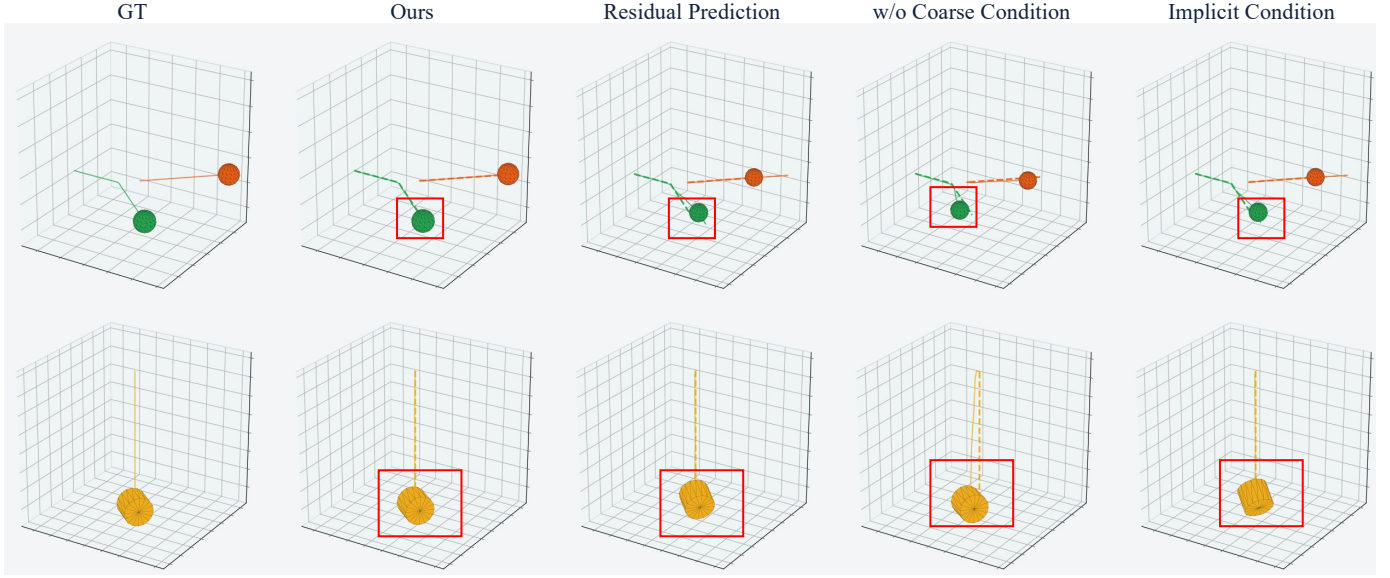


Fig. 8. **Ablation of motion-plan conditioning in the dense decoder.** Each row shows one scene, comparing direct trajectory prediction conditioned on the motion plan (Ours) with residual prediction, no motion plan, and implicit VLM conditioning. Direct prediction with the explicit plan most closely follows the ground-truth paths and final poses, while the alternatives introduce trajectory or orientation deviations.

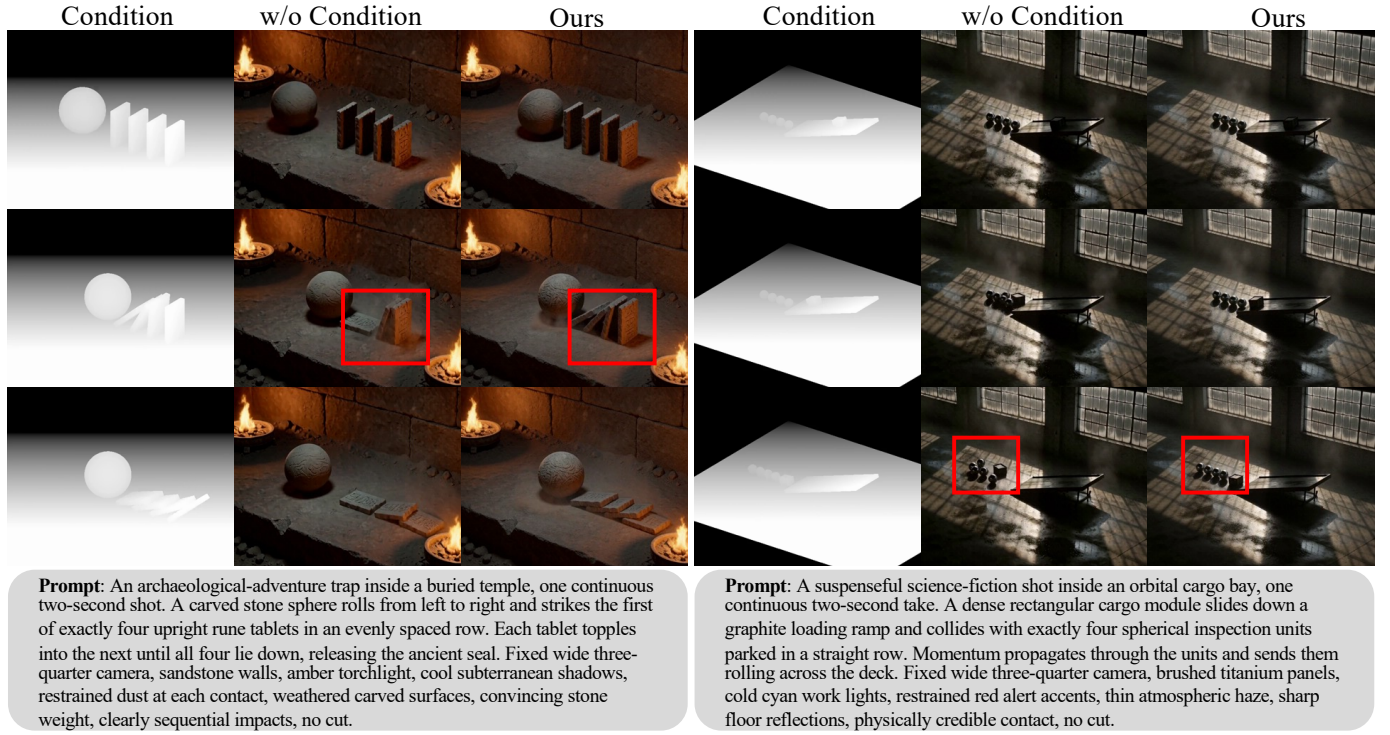


Fig. 9. **Application to physics-guided video generation.** Our predicted trajectories are rendered as depth videos to spatially condition MiniMax-H3. For each scene, columns show the depth condition, unguided generation (w/o Condition), and generation with our trajectory guidance; rows show successive frames. In the w/o Condition results, the four upright tablets become three in the left scene, while the four spheres become five in the right scene. Guided results better preserve object counts and collision-driven motion. Red boxes highlight differences in object persistence and interaction outcomes.

learning-based simulator such as PhysiFormer. Nevertheless, our complete pipeline is faster than the API-based Gemini 3.0 baseline, although the measurements use different hardware and are therefore not directly comparable.

VI. DISCUSSION

A. Conclusion

We presented a reason, plan, and animate framework for learning-based rigid-body motion forecasting that connects physical event reasoning, sparse motion planning, and dense trajectory generation. A trajectory-grounded VLM predicts an object-centric event trace and sparse translation waypoints,

TABLE VII

INFERENCE LATENCY PER SAMPLE (SECONDS). PLANNING INCLUDES PHYSICAL EVENT REASONING AND MOTION PLAN GENERATION, WHILE DECODING DENOTES DENSE TRAJECTORY PREDICTION. [†]GEMINI 3.0 IS MEASURED THROUGH ITS API ON DIFFERENT HARDWARE.

Method	Planning	Decoding	Total
PhysiFormer	–	0.41	0.41
Gemini 3.0 [†]	4.84	–	4.84
Ours	2.51	0.33	2.84

while a shape-aware flow-matching decoder generates dense SE(3) trajectories conditioned on the plan and initial object state. The explicit numerical plan makes the interface between these stages inspectable and editable, without requiring the VLM to produce dense poses or the decoder to interpret language-model hidden states.

Experiments on MOVi and RIGID+ support both event supervision for waypoint prediction and explicit plan conditioning for dense reconstruction. Under the tested distribution shifts, the complete model attains lower translation and rotation errors than PhysiFormer when averaged across OOD axes, with a small translation disadvantage under the object-count shift. Qualitative video-generation examples further illustrate the reuse of these trajectories as depth guidance, separating motion control from appearance synthesis. Together, these findings support using VLMs for trajectory-grounded event reasoning and motion planning while delegating fine-grained trajectory generation to a geometry-aware decoder.

B. Limitations and future work

First, our framework uses supplied object geometry, initial states, and available physical parameters, which are not directly given in real-world observations. Extending it to real-world rigid-body motion forecasting requires additional modules to estimate physical parameters, geometry, and initial states.

Second, the current framework does not explicitly enforce physical constraints such as non-penetration, conservation laws, and contact constraints. As a result, a predicted trajectory may look reasonable while still violating basic physical rules. Future work could add physics-based training objectives and use reinforcement learning with physical feedback to further post-train the model. This would allow physically invalid predictions to be directly penalized and improve the physical validity of the generated trajectories.

REFERENCES

- [1] T. Brooks, B. Peebles, C. Holmes, W. DePue, Y. Guo, L. Jing, D. Schnurr, J. Taylor, T. Luhman, E. Luhman, C. Ng, R. Wang, and A. Ramesh, “Video generation models as world simulators,” OpenAI technical report, 2024. [Online]. Available: <https://openai.com/index/video-generation-models-as-world-simulators/>
- [2] T. Yuan, Z. Dong, Y. Liu, and H. Zhao, “Fast-wam: Do world action models need test-time future imagination?” *arXiv preprint arXiv:2603.16666*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.16666>
- [3] L. Li, Q. Zhang, Y. Luo, S. Yang, R. Wang, F. Han, M. Yu, Z. Gao, N. Xue, X. Zhu, Y. Shen, and Y. Xu, “Causal world modeling for robot control,” *arXiv preprint arXiv:2601.21998*, 2026. [Online]. Available: <https://arxiv.org/abs/2601.21998>
- [4] A. Hu, L. Russell, H. Yeo, Z. Murez, G. Fedoseev, A. Kendall, J. Shotton, and G. Corrado, “GAIA-1: A generative world model for autonomous driving,” *arXiv preprint arXiv:2309.17080*, 2023. [Online]. Available: <https://arxiv.org/abs/2309.17080>
- [5] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, “Self-supervised learning from images with a joint-embedding predictive architecture,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 15 619–15 629.
- [6] M. Assran, A. Bardes, D. Fan, Q. Garrido, R. Howes, M. Komeili, M. Muckley, A. Rizvi, C. Roberts, K. Sinha, A. Zholus, S. Arnaud, A. Gejji, A. Martin, F. R. Hogan, D. Dugas, P. Bojanowski, V. Khalidov, P. Labatut, F. Massa, M. Szafraniec, K. Krishnakumar, Y. Li, X. Ma, S. Chandar, F. Meier, Y. LeCun, M. Rabbat, and N. Ballas, “V-JEPA 2: Self-supervised video models enable understanding, prediction and planning,” 2025. [Online]. Available: <https://arxiv.org/abs/2506.09985>
- [7] T. Wiedemer, Y. Li, P. Vicol, S. S. Gu, N. Matarese, K. Swersky, B. Kim, P. Jaini, and R. Geirhos, “Video models are zero-shot learners and reasoners,” 2025. [Online]. Available: <https://arxiv.org/abs/2509.20328>
- [8] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851.
- [9] W. Peebles and S. Xie, “Scalable diffusion models with transformers,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4195–4205.
- [10] X. Huang, Z. Li, G. He, M. Zhou, and E. Shechtman, “Self forcing: Bridging the train-test gap in autoregressive video diffusion,” in *Advances in Neural Information Processing Systems*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.08009>
- [11] H. Zhu, M. Zhao, G. He, H. Su, C. Li, and J. Zhu, “Causal forcing: Autoregressive diffusion distillation done right for high-quality real-time interactive video generation,” in *International Conference on Machine Learning*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.02214>
- [12] A. Sanchez-Gonzalez, J. Godwin, T. Pfaff, R. Ying, J. Leskovec, and P. W. Battaglia, “Learning to simulate complex physics with graph networks,” in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119. PMLR, 2020, pp. 8459–8468.
- [13] K. R. Allen, Y. Rubanova, T. Lopez-Guevara, W. Whitney, A. Sanchez-Gonzalez, P. Battaglia, and T. Pfaff, “Learning rigid dynamics with face interaction graph networks,” in *The Eleventh International Conference on Learning Representations*, 2023. [Online]. Available: <https://arxiv.org/abs/2212.03574>
- [14] Y. Rubanova, T. Lopez-Guevara, K. R. Allen, W. F. Whitney, K. Stachenfeld, and T. Pfaff, “Learning rigid-body simulators over implicit shapes for large-scale scenes and vision,” in *Advances in Neural Information Processing Systems*, A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, Eds., vol. 37. Curran Associates, Inc., 2024, pp. 125 809–125 838.
- [15] Y. Shao, C. C. Loy, and B. Dai, “Transformer with implicit edges for particle-based physics simulation,” in *Computer Vision – ECCV 2022*. Springer Nature Switzerland, 2022, pp. 549–564.
- [16] C. Wang, C. Chen, Y. Huang, Z. Dou, Y. Liu, J. Gu, and L. Liu, “Physctrl: Generative physics for controllable and physics-grounded video generation,” in *Advances in Neural Information Processing Systems 38*. Neural Information Processing Systems Foundation, Inc., 2025, pp. 186 107–186 132.
- [17] Y. Chen, Y. Lan, and A. Vedaldi, “Physiformer: Learning to simulate mechanics in world space,” 2026. [Online]. Available: <https://arxiv.org/abs/2606.27364>
- [18] Y. Hu, Y. Fang, Z. Ge, Z. Qu, Y. Zhu, A. Pradhana, and C. Jiang, “A moving least squares material point method with displacement discontinuity and two-way rigid body coupling,” *ACM Transactions on Graphics (TOG)*, vol. 37, no. 4, pp. 1–14, 2018.
- [19] K. Greff, F. Belletti, L. Beyer, C. Doersch, Y. Du, D. Duckworth, D. J. Fleet, D. Gnanapragasam, F. Golemo, C. Herrmann, T. Kipf, A. Kundu, D. Lagun, I. Laradji, H.-T. D. Liu, H. Meyer, Y. Miao, D. Nowrouzezahrai, C. Ozireli, E. Pot, N. Radwan, D. Rebain, S. Sabour, M. S. M. Sajjadi, M. Sela, V. Sitzmann, A. Stone, D. Sun, S. Vora, Z. Wang, T. Wu, K. M. Yi, F. Zhong, and A. Tagliasacchi, “Kubric: A scalable dataset generator,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 3749–3761.
- [20] G. Comanici, E. Bieber, M. Schaeckermann, I. Pasupat, N. Sachdeva, I. Dhillon, M. Blistein, O. Ram, D. Zhang, E. Rosen *et al.*, “Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities,” *arXiv preprint*

- arXiv:2507.06261, 2025. [Online]. Available: <https://arxiv.org/abs/2507.06261>
- [21] Anthropic, “Claude fable 5,” June 2026, large language model; launched June 9, 2026; accessed September 24, 2026. [Online]. Available: <https://www.anthropic.com/news/claude-fable-5-mythos-5>
 - [22] X. Yang, B. Li, Y. Zhang, Z. Yin, L. Bai, L. Ma, Z. Wang, J. Cai, T.-T. Wong, H. Lu, and X. Jia, “VLIPP: Towards physically plausible video generation with vision and language informed physical prior,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2025, pp. 12 360–12 370.
 - [23] H.-Y. Lee, H. Jiang, Y. Mei, J. Shi, M.-H. Yang, and Z. Shu, “Motion: Motion-controlled video generation with visual reasoning,” in *Forty-third International Conference on Machine Learning*, 2026.
 - [24] M. B. Kizil, E. Sanli, N. J. Mitra, E. Erdem, A. Erdem, and D. Ceylan, “LAMP: Language-assisted motion planning for controllable video generation,” pp. 12 829–12 838, June 2026.
 - [25] T. Pfaff, M. Fortunato, A. Sanchez-Gonzalez, and P. W. Battaglia, “Learning mesh-based simulation with graph networks,” in *The Ninth International Conference on Learning Representations*, 2021.
 - [26] A. Wei and O. Fink, “Integrating physics and topology in neural networks for learning rigid body dynamics,” *Nature Communications*, vol. 16, no. 1, p. 6867, 2025.
 - [27] Y.-Y. Yu, J. Choi, W. Cho, K. Lee, N. Kim, K. Chang, C. Woo, I. KIM, S. Lee, J. Y. Yang, S. YOON, and N. Park, “Learning flexible body collision dynamics with hierarchical contact mesh transformer,” in *The Twelfth International Conference on Learning Representations*, 2024.
 - [28] W. F. Whitney, J. Varley, D. Jain, K. Choromanski, S. Singh, and V. Sindhwani, “Modeling the real world with high-density visual particle dynamics,” 2024. [Online]. Available: <https://arxiv.org/abs/2406.19800>
 - [29] J. Yim, B. L. Trippe, V. De Bortoli, E. Mathieu, A. Doucet, R. Barzilay, and T. Jaakkola, “SE(3) diffusion model with application to protein backbone generation,” in *International Conference on Machine Learning*, 2023.
 - [30] J. Urain, N. Funk, J. Peters, and G. Chalvatzaki, “SE(3)-DiffusionFields: Learning smooth cost functions for joint grasp and motion optimization through diffusion,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 5923–5930.
 - [31] C.-C. Hsu, B. Wen, J. Xu, Y. Narang, X. Wang, Y. Zhu, J. Biswas, and S. Birchfield, “SPOT: SE(3) pose trajectory diffusion for object-centric manipulation,” 2024. [Online]. Available: <https://arxiv.org/abs/2411.00965>
 - [32] P. W. Battaglia, R. Pascanu, M. Lai, D. Jimenez Rezende, and K. Kavukcuoglu, “Interaction networks for learning about objects, relations and physics,” in *Advances in Neural Information Processing Systems*, vol. 29. Curran Associates, Inc., 2016, pp. 4502–4510.
 - [33] M. B. Chang, T. Ullman, A. Torralba, and J. B. Tenenbaum, “A compositional object-based approach to learning physical dynamics,” in *International Conference on Learning Representations*, 2017.
 - [34] Y. Li, J. Wu, R. Tedrake, J. B. Tenenbaum, and A. Torralba, “Learning particle dynamics for manipulating rigid bodies, deformable objects, and fluids,” in *International Conference on Learning Representations*, 2019.
 - [35] Z. Dou, M. Guo, H. Wu, D. Roble, T. Stuyck, and W. Matusik, “Rigid-former: Learning rigid dynamics using transformers,” *arXiv preprint arXiv:2605.09196*, 2026.
 - [36] C. Wang, M. Guo, S. Chen, H. Zhang, M. Wang, X. Ni, H. Sun, K. Wang, Z. Pan, K. Wu, L. Liu, Y. Yang, C. Jiang, T. Komura, W. Matusik, and P. Y. Chen, “WorldParticle: Unified world simulation of lagrangian particle dynamics via transformer,” 2026. [Online]. Available: <https://arxiv.org/abs/2605.15305>
 - [37] D. Guo *et al.*, “DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning,” *Nature*, vol. 645, no. 8081, pp. 633–638, September 2025.
 - [38] X. Chen, R. Zhang, D. Jiang, A. Zhou, S. Yan, W. Lin, and H. Li, “Mintcot: Enabling interleaved visual tokens in mathematical chain-of-thought reasoning,” in *Advances in Neural Information Processing Systems* 38, 2025, pp. 77 061–77 090.
 - [39] Z. Huang, N. Yu, G. Chen, H. Qiu, P. Debevec, and Z. Liu, “VChain: Chain-of-visual-thought for reasoning in video generation,” in *Findings of the Association for Computational Linguistics: ACL 2026*, M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, Eds. San Diego, California, United States: Association for Computational Linguistics, July 2026, pp. 226–250.
 - [40] D. Jiang, Z. Guo, R. Zhang, Z. Zong, H. Li, L. Zhuo, S. Yan, P.-A. Heng, and H. Li, “T2i-r1: Reinforcing image generation with collaborative semantic-level and token-level cot,” in *Advances in Neural Information Processing Systems* 38, 2025, pp. 44 581–44 615.
 - [41] Z. Guo, R. Zhang, H. Li, M. Zhang, X. Chen, S. Wang, Y. Feng, P. Pei, and P.-A. Heng, “Thinking-while-generating: Interleaving textual reasoning throughout visual generation,” *arXiv:2511.16671*, 2025.
 - [42] H. Li, T. Ren, X. Ma, C. Qing, Z. Fang, S. He, Z. Guo, H. Wu, J. Tian, Y. Zou, R. An, D. Jiang, B. Yang, J. Xie, X. Huang, W. Yan, J. Zou, Z. Yue, Y. Luo, X. Li, Y. Wang, J. Ye, J. Zhao, Z. Chen, L. Chen, R. Yan, F. Zhao, and P.-A. Heng, “Videococo: Code-as-cot for physically-consistent video generation via an agentic dual-engine system,” 2026. [Online]. Available: <https://arxiv.org/abs/2607.27380>
 - [43] J. Wang, A. Ma, K. Cao, J. Zheng, Z. Zhang, J. Feng, S. Liu, Y. Ma, B. Cheng, D. Leng, Y. Yin, and X. Liang, “Wisa: World simulator assistant for physics-aware text-to-video generation,” 2025. [Online]. Available: <https://arxiv.org/abs/2503.08153>
 - [44] J. Zhang, C. Zheng, Y. Yang, M. Argus, R. Soraki, W. Han, T. Anderson, C.-L. Li, S. Liu, J. Duan, Z. Ren, J. Zhang, and R. Krishna, “Molmotation: Forecasting point trajectories in 3d with language instruction,” 2026. [Online]. Available: <https://arxiv.org/abs/2606.18558>
 - [45] H. Wang, H. Ouyang, Q. Wang, W. Wang, Q. Bai, K. L. Cheng, Y. Yu, Y. Li, Y. Meng, Z. Liu, Y. Zeng, Y. Shen, and Q. Chen, “Worlddirector: Building controllable world simulators with persistent dynamic memory,” *arXiv preprint arXiv:2607.02517*, 2026.
 - [46] A. J. Hancock, X. Wu, L. Zha, O. Russakovsky, and A. Majumdar, “Actions as language: Fine-tuning vlms into vlms without catastrophic forgetting,” 2025. [Online]. Available: <https://arxiv.org/abs/2509.22195>
 - [47] Z. Zhao, W. Liu, X. Chen, X. Zeng, R. Wang, P. Cheng, B. FU, T. Chen, G. YU, and S. Gao, “Michelangelo: Conditional 3d shape generation based on shape-image-text aligned latent representation,” in *Thirty-seventh Conference on Neural Information Processing Systems*, 2023, pp. 73 969–73 982.
 - [48] Physical Intelligence, K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, J. Tanner, Q. Vuong, H. Walke, A. Walling, H. Wang, L. Yu, and U. Zhilinsky, “ $\pi_{0.5}$: a vision-language-action model with open-world generalization,” 2025. [Online]. Available: <https://arxiv.org/abs/2504.16054>
 - [49] S. Wei, H. Jing, B. Li, Z. Zhao, J. Mao, Z. Ni, S. He, S. Zang, X. Liu, K. Kang, J. Liu, W. Yuan, M. Pavone, D. Huang, and Y. Wang, “ Ψ_0 : An open foundation model towards universal humanoid loco-manipulation,” in *Proceedings of Robotics: Science and Systems*, Sydney, Australia, July 2026.
 - [50] C. Mou, X. Wang, L. Xie, Y. Wu, J. Zhang, Z. Qi, Y. Shan, and X. Qie, “T2I-Adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models,” 2023. [Online]. Available: <https://arxiv.org/abs/2302.08453>
 - [51] L. Zhang, A. Rao, and M. Agrawala, “Adding conditional control to text-to-image diffusion models,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2023, pp. 3836–3847.
 - [52] J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, and Y. Liu, “RoFormer: Enhanced transformer with rotary position embedding,” *Neurocomputing*, vol. 568, p. 127063, February 2024.
 - [53] T. Li and K. He, “Back to basics: Let denoising generative models denoise,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 36 115–36 125.
 - [54] E. Coumans and Y. Bai, “Pybullet, a python module for physics simulation for games, robotics and machine learning,” <http://pybullet.org>, 2016–2021.
 - [55] Qwen Team, “Qwen3.5: Towards native multimodal agents,” February 2026, official Qwen blog; published 16 February 2026. [Online]. Available: <https://qwen.ai/blog?id=qwen3.5>
 - [56] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “Lora: Low-rank adaptation of large language models,” in *The Tenth International Conference on Learning Representations*, 2022.
 - [57] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *7th International Conference on Learning Representations (ICLR)*, 2019.
 - [58] P. Esser, S. Kulal, A. Blattmann, R. Entezari, J. Müller, H. Saini, Y. Levi, D. Lorenz, A. Sauer, F. Boesel, D. Podell, T. Dockhorn, Z. English, and R. Rombach, “Scaling rectified flow transformers for high-resolution image synthesis,” in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 12 606–12 633.

- [59] K. Shoemake, "Animating rotation with quaternion curves," in *Proceedings of the 12th annual conference on Computer graphics and interactive techniques*, ser. SIGGRAPH '85. ACM, 1985, pp. 245–254.
- [60] K. Pertsch, K. Stachowicz, B. Ichter, D. Driess, S. Nair, Q. Vuong, O. Mees, C. Finn, and S. Levine, "FAST: Efficient action tokenization for vision-language-action models," in *Robotics: Science and Systems XXI*, 2025.
- [61] Z. Gu, R. Yan, J. Lu, P. Li, Z. Dou, C. Si, Z. Dong, Q. Liu, C. Lin, Z. Liu, W. Wang, and Y. Liu, "Diffusion as shader: 3d-aware video diffusion for versatile video generation control," *arXiv preprint arXiv:2501.03847*, 2025.
- [62] MiniMax-AI, "MiniMax H3," GitHub repository, 2026. [Online]. Available: <https://github.com/MiniMax-AI/MiniMax-H3>